

The $z_1 \rightarrow x_{\text{pom}}$ Jacobian in the Exclusive Cross Section

Exclusive kinematics

The exclusive calculation is differential in the observed rapidity y and transverse momentum, and integrates directly over the photon plus-momentum q^+ . The charm-quark plus-momentum is fixed by y :

$$p^+ = \frac{m_T}{\sqrt{2}} e^y, \quad z_1 = \frac{p^+}{q^+}, \quad z_2 = 1 - z_1, \quad m_T = \sqrt{p_c^2 + m_c^2}. \quad (1)$$

The code therefore uses dq^+ as the integration measure. It computes x_{pom} from the sampled q^+ , but does not sample x_{pom} as an independent integration variable.

Using $W^2 = \sqrt{2} q^+ s$ and $q^+ = p^+/z_1$, one obtains

$$x_{\text{pom}} = \frac{M_{c\bar{c}}^2}{W^2} = \frac{p_c^2 + m_c^2}{W^2 z_1 z_2} = \frac{m_T}{s e^y z_2} = \frac{m_T}{s e^y (1 - z_1)}. \quad (2)$$

Thus, at fixed y and p_c , x_{pom} is a monotonic function of z_1 . There are no two-root ambiguities in this exclusive kinematics.

Jacobian if z_1 is replaced by x_{pom}

Differentiating the exclusive relation gives

$$\frac{dx_{\text{pom}}}{dz_1} = \frac{x_{\text{pom}}}{z_2}, \quad \boxed{\left| \frac{dz_1}{dx_{\text{pom}}} \right| = \frac{z_2}{x_{\text{pom}}} = \frac{1 - z_1}{x_{\text{pom}}}}. \quad (3)$$

Equivalently, insertion of a delta function gives

$$\delta(x_{\text{pom}} - X(z_1)) = \frac{z_2}{x_{\text{pom}}} \delta(z_1 - z_1(x_{\text{pom}})). \quad (4)$$

Therefore,

$$\frac{d\sigma}{dx_{\text{pom}}} = \left. \frac{d\sigma}{dz_1} \right|_{z_1=z_1(x_{\text{pom}})} \frac{1 - z_1}{x_{\text{pom}}}. \quad (5)$$

Relation to the implemented calculation

The exclusive Monte Carlo integral is proportional to

$$\int db dq^+ dz_h z_1 m_c^2 g(p_c, z_1, x_{\text{pom}}) \Phi(b, q^+) \frac{D_{c \rightarrow D^0}(z_h)}{z_h^2}, \quad (6)$$

with $p_c = p_{D^0}/z_h$. Since the integration variable is q^+ , the Jacobian above is not an extra factor in the code. If the integration is rewritten from q^+ to x_{pom} , the required measure is

$$\left| \frac{dq^+}{dx_{\text{pom}}} \right| dx_{\text{pom}} = \frac{q^+ z_2}{x_{\text{pom}} z_1} dx_{\text{pom}}. \quad (7)$$

The $z_1 \rightarrow x_{\text{pom}}$ form follows from $|dz_1/dx_{\text{pom}}| = z_2/x_{\text{pom}}$.

The often-quoted expression $z_1 z_2/[x_{\text{pom}}|1 - 2z_1|]$ assumes that W is held fixed while z_1 is varied. That is not the exclusive UPC integration used here, because $W^2 = \sqrt{2} q^+ s$ changes with the integration variable q^+ .

Practical conclusion

The plotted observable in the exclusive cross-section PDF is $d\sigma/(dy dp_{D^0\perp})$. The code integrates over q^+ and only uses x_{pom} inside the dipole amplitude and the condition $0 < x_{\text{pom}} \leq 0.1$. Therefore no $z_1 \rightarrow x_{\text{pom}}$ Jacobian should be multiplied into the existing exclusive integrand.